

D771

Quantitative Literacy

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A teacher deposits \$2,000 in a savings account that pays 5% interest compounded annually. The bank has a branch next to her school and offers free checks. How much will the account hold after 3 years if she makes no other deposits?

- A. \$2,300.00
- B. \$2,315.25
- C. \$2,100.00
- D. \$6,300.00

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A loan of \$1,500 charges simple interest. After 3 years, the total interest is \$180. What is the annual interest rate?

- A. 12%
- B. 36%
- C. 18%
- D. 4%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A store raises the price of a \$50 jacket by 20% in the fall, then puts it on sale for 20% off in the spring. The jacket comes in four colors and is made of recycled material. What is the spring sale price?

- A. \$50
- B. \$60
- C. \$48
- D. \$40

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

After a 15% discount, a pair of shoes costs \$68. What was the original price?

- A.** \$57.80
- B.** \$80
- C.** \$83
- D.** \$78.20

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A car rental costs $C = 25 + 0.40m$ dollars, where m is miles driven. The company has a 4.5-star rating and offers a free car wash. What does the 0.40 represent?

- A.** The cost for each additional mile driven.
- B.** The total cost of the rental.
- C.** The value of the free car wash.
- D.** The flat fee charged before any miles.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Using $C = 25 + 0.40m$, a customer's bill is \$73. How many miles did she drive?

- A.** 182.5 miles
- B.** 245 miles
- C.** 120 miles
- D.** 19.2 miles

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Town A has 10,000 people and grows by 300 people a year. Town B has 10,000 people and grows 3% a year. Town B has a new library and a lake. Which statement is true?

- A.** The towns grow the same amount every year forever.
- B.** Town B grows faster because of its library and lake.
- C.** Town A grows faster because 300 is larger than 3.
- D.** The towns are equal after one year, but Town B pulls ahead after that because its growth compounds.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A 12-ounce jar of peanut butter costs \$3.00. A 20-ounce jar costs \$4.40. The smaller jar has a colorful label and is on an eye-level shelf. Which is the better buy per ounce?

- A. The 12-ounce jar, at \$0.22 per ounce.
- B. The 12-ounce jar, because it costs less.
- C. The 20-ounce jar, at \$0.22 per ounce.
- D. They are equal.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A school garden is 10 feet by 6 feet. The principal wants a new garden twice as long and twice as wide, near the new gym. How does the new area compare?

- A. 240 square feet, four times the original.
- B. 120 square feet, twice the original.
- C. 64 square feet.
- D. 60 square feet.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A raised garden bed is 8 feet long, 4 feet wide, and 1.5 feet deep. Soil comes in 2-cubic-foot bags that cost \$6 each at a store that opens at 7 a.m. How many bags are needed to fill it?

- A. 24 bags
- B. 16 bags
- C. 8 bags
- D. 48 bags

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A city's unemployment rate rose from 8% to 10%. A news anchor with 20 years of experience reported it on the evening news. Which statement is accurate?

- A. It rose 18 percentage points.
- B. It rose 2%.
- C. It rose 20%.
- D. It rose 2 percentage points, a 25% increase.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A graduation rate increased by 10% from a starting rate of 80%. What is the new rate?

- A.** 8%
- B.** 88%
- C.** 90%
- D.** 72%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A teacher deposits \$2,000 in a savings account that pays 5% interest compounded annually. ~~The bank has a branch next to her school and offers free checks.~~ How much will the account hold after 3 years if she makes no other deposits?

Target word: compound amount **Decides:** $A = 2000 \times 1.05^3$.

A	Adjacent category	Simple interest: $2,000 \times 0.05 \times 3 = 300$ added.
B	Correct	$2,000 \times 1.157625 = 2,315.25$.
C	Dropped element	Only one year of interest.
D	Inverted operation	Multiplies $2,000 \times 1.05$ by 3 instead of raising 1.05 to the third power.

2. Answer D Concept 1 • transfer

A loan of \$1,500 charges simple interest. After 3 years, the total interest is \$180. What is the annual interest rate?

Target word: solve for rate **Decides:** $r = I / (P \times t) = 180 / 4,500$.

A	Dropped element	$180 / 1,500$ ignores the 3 years.
B	Inverted operation	Multiplies by 3 instead of dividing.
C	Wrong stem number	Reads the \$180 as a rate.
D	Correct	$180 / (1,500 \times 3) = 0.04$.

3. Answer C Concept 2

A store raises the price of a \$50 jacket by 20% in the fall, then puts it on sale for 20% off in the spring. ~~The jacket comes in four colors and is made of recycled material.~~ What is the spring sale price?

Target word: price after two changes **Decides:** $50 \times 1.2 = 60$; $60 \times 0.8 = 48$.

A	Inverted operation	Assumes equal percent changes cancel.
B	Wrong stem number	The fall price, before the sale.
C	Correct	The 20% cut applies to \$60, removing \$12.
D	Dropped element	Applies the 20% cut to the original \$50, dropping the increase.

4. Answer B Concept 2 • transfer

After a 15% discount, a pair of shoes costs \$68. What was the original price?

Target word: solve for original **Decides:** $\text{Original} \times 0.85 = 68$.

A	Confusion set	Takes another 15% off.
B	Correct	$68 / 0.85 = 80$.
C	Wrong stem number	Adds 15 dollars instead of 15 percent.
D	Inverted operation	Adds 15% of \$68 instead of dividing by 0.85.

5. Answer A Concept 3

A car rental costs $C = 25 + 0.40m$ dollars, where m is miles driven. ~~The company has a 4.5 star rating and offers a free car wash.~~ What does the 0.40 represent?

Target word: meaning of slope **Decides:** 0.40 is the coefficient of m .

A	Correct	The slope is the change in cost per mile.
B	Adjacent category	Total cost is C , not a parameter.
C	Camouflage	The car wash is irrelevant.
D	Inverted operation	That is the intercept, \$25.

6. Answer C Concept 3 • transfer

Using $C = 25 + 0.40m$, a customer's bill is \$73. How many miles did she drive?

Target word: solve for x **Decides:** $(73 - 25) / 0.40 = 120$.

A	Dropped element	$73 / 0.40$ ignores the \$25 fee.
B	Inverted operation	Adds the fee: $(73 + 25) / 0.40$.
C	Correct	$48 / 0.40 = 120$.
D	Confusion set	Multiplies 48 by 0.40 instead of dividing.

7. Answer D Concept 4

Town A has 10,000 people and grows by 300 people a year. Town B has 10,000 people and grows 3% a year. ~~Town B has a new library and a lake.~~ Which statement is true?

Target word: linear vs. exponential **Decides:** Town B multiplies each year; Town A adds a constant.

A	Absolute language	3% of a growing population is more than 300 after year 1.
B	Camouflage	Amenities do not decide the math.
C	Confusion set	Compares a count to a percent.
D	Correct	3% of 10,000 = 300 in year 1; later, 3% applies to a larger base.

8. Answer C Concept 5

A 12-ounce jar of peanut butter costs \$3.00. A 20-ounce jar costs \$4.40. ~~The smaller jar has a colorful label and is on an eye-level shelf.~~ Which is the better buy per ounce?

Target word: unit price **Decides:** $3.00 / 12 = \$0.25$; $4.40 / 20 = \$0.22$.

A	Confusion set	Attaches the right unit price to the wrong jar.
B	Dropped element	Compares total price, not price per ounce.
C	Correct	\$0.22 is less than \$0.25 per ounce.
D	Inverted operation	The unit prices differ.

9. Answer A Concept 6

A school garden is 10 feet by 6 feet. The principal wants a new garden twice as long and twice as wide, ~~near the new gym~~. How does the new area compare?

Target word: area scaling **Decides:** Scale factor 2 → area × 4.

A	Correct	$20 \times 12 = 240 = 4 \times 60$.
B	Inverted operation	Doubles area instead of squaring the scale factor.
C	Adjacent category	That is the new perimeter, $2 \times (20 + 12)$, not area.
D	Wrong stem number	The original area.

10. Answer A Concept 7

A raised garden bed is 8 feet long, 4 feet wide, and 1.5 feet deep. Soil comes in 2-cubic-foot bags ~~that cost \$6 each at a store that opens at 7 a.m.~~ How many bags are needed to fill it?

Target word: volume then containers **Decides:** $8 \times 4 \times 1.5 = 48$ cubic feet; $48 / 2 = 24$.

A	Correct	48 cubic feet / 2 per bag = 24.
B	Adjacent category	Uses area 32 square feet / 2, ignoring depth.
C	Camouflage	Divides 48 by the \$6 price.
D	Dropped element	Drops the division by bag size.

11. Answer D Concept 8

A city's unemployment rate rose from 8% to 10%. ~~A news anchor with 20 years of experience reported it on the evening news.~~ Which statement is accurate?

Target word: points vs. percent **Decides:** 2 points; $2 / 8 = 25\%$ relative increase.

A	Wrong stem number	Adds the two rates.
B	Confusion set	Confuses percentage points with percent change.
C	Inverted operation	Divides 2 by 10 instead of the starting 8.
D	Correct	$10 - 8 = 2$ points; $2 / 8 = 0.25$.

12. Answer B Concept 8 · transfer

A graduation rate increased by 10% from a starting rate of 80%. What is the new rate?

Target word: apply a relative percent change **Decides:** $80 \times 1.10 = 88$.

A	Dropped element	The change only, not the new rate.
B	Correct	A 10% relative increase on 80 is 8 points.
C	Confusion set	Adds 10 percentage points instead of 10 percent.
D	Inverted operation	Decreases instead of increases.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Simple vs. compound interest	Simple interest = $P \times r \times t$ on the original principal only; compound interest earns interest on past interest: $A = P(1 + r)^t$ for annual compounding.	Using simple interest when the problem says compounded, or the reverse.	“Compounded annually” vs. “simple interest.”
2	Percent change is not symmetric	Each percent change applies to the current value, so a 20% increase followed by a 20% decrease leaves you 4% below where you started.	Assuming equal percent ups and downs cancel out.	Two percent changes in a row, or undoing a discount.
3	Reading a linear model	In $y = b + mx$, b is the starting value (when $x = 0$) and m is the constant change in y for each 1-unit increase in x .	Swapping the slope and the intercept.	A flat fee plus a per-unit rate.
4	Linear vs. exponential growth	Linear growth adds the same amount each period; exponential growth multiplies by the same factor each period, so it pulls ahead over time.	Treating “grows 3% a year” as adding the same number every year.	A fixed amount per period vs. a fixed percent per period.
5	Unit rates for comparison	Compare options by converting to the same unit (cost per ounce, miles per gallon) rather than by total price.	Choosing the package with the lower sticker price.	Two sizes with two prices.
6	Scaling area and volume	If lengths scale by k , area scales by k^2 and volume by k^3 .	Assuming doubling the sides doubles the area.	“Twice as long and twice as wide.”
7	Volume of a rectangular prism	Volume = length $\times$ width $\times$ height, in cubic units; divide by the size of one container to find how many are needed.	Finding area (two dimensions) when volume is asked, or mixing units.	Filling a box, bed, or tank.
8	Percentage points vs. percent	A change from 8% to 10% is 2 percentage points but a 25% relative increase.	Calling a 2-point rise a 2% increase.	Two rates given as percents and a question about the change.

Course scope (public description): Quantitative Literacy views real-world problems through the lens of quantitative reasoning. The application of quantitative concepts to topics such as financial decisions is explored. Algebraic models and functions, as well as principles of geometry, are reviewed as fundamental ways to explore real-life scenarios. The use of mathematical concepts as a tool for modeling and understanding everyday problems is leveraged to promote students' thinking of math as a useful and relevant tool for many situations and scenarios. Numeracy and quantitative thinking skills are developed through these applications.